Nishchal Sapkota

Machine Learning Scientist specializing in vision foundation models, self-supervised and multimodal learning, and healthcare AI

✉ nsapkota@nd.edu linkedin.com/in/nsapkota417 Personal Website Google Scholar

Education

University of Notre Dame (UND) Notre Dame, IN
Ph.D. *in Computer Science and Engineering* *12/2026*
M.S. *in Computer Science and Engineering* *08/2024*
Research Areas: ***Computer Vision, Self-supervised Learning, Foundation Models, Data-efficient Deep Learning Models, AI for Healthcare***

The University of Southern Mississippi (USM) Hattiesburg, MS
B.S. *in* ***Computer Science*** | ***B.S.*** *in* ***Mathematics*** *– Honors, summa cum laude (GPA: 3.91)* *08/2020*
Thesis: Probabilistic Analysis of Revenues in Online Games

Experiences

Mayo Clinic Rochester, MN
Computational Pathology and AI Intern *01/2025 – 07/2026*

- Developed the ***largest pathology vision foundation model*** with 300× less data and achieving up to 5% improvement over state-of-the-art methods through scaling-aware training and mid-training regularization to resolve scaling bottlenecks. [1]
- Assisted in extending pathology foundation models into ***multimodal vision-language models*** by integrating patient diagnosis reports with whole-slide pathology images.
- Built large-scale healthcare AI systems using ***distributed training*** and scalable ***MLOps pipelines*** for clinical predictive modeling. [9]

IBM Durham, NC
Senior Data Science & AI PhD Intern *05/2025 – 08/2025*

- Developed **user-behavior and session-level predictive models** (N-gram Markov, Transformer, Mamba) on large-scale web clickstream data to identify high-intent users and predict conversion outcomes, achieving a 14% improvement over internal baselines.
- Deployed models to enable real-time agentic AI marketing interventions, recovering up to **10,000 lost conversion opportunities per quarter**, while collaborating cross-functionally to translate model insights into business strategy.

The University of Notre Dame Notre Dame, IN
Graduate Researcher *08/2020 – Present*

- Designed a memory-augmented ***video segmentation model*** [2] to advance AI-assisted surgery, outperforming foundational models (SAM2/3) in public surgical benchmarks by up to 17% and developed lightweight variants for real-time deployment.
- Pioneered three data-efficient ***3D segmentation models*** with up to 11% improvement on medical image/volume segmentation benchmarks. [3][4][5]
- Built ***self-supervised learning*** frameworks achieving up to 4% gains across medical image segmentation benchmarks. [10][11][12]
- Proposed ***multimodal learning*** frameworks and shape-aware segmentation/classification models using ***implicit neural representations*** by handling label ambiguity and improving data efficiency by 30%. [6][13][16]
- ***Collaborated with academic, clinical, and industry labs*** to develop AI-driven healthcare solutions and ***mentored 5+ students***, resulting in multiple publications and successful industry and academic placements. [6][14][15][17][18]

The University of Southern Mississippi Hattiesburg, MS
Undergraduate Researcher *08/2017 – 05/2020*

- Analyzed online games using Markov Chain to ***maximize revenues*** for both players and the providers. [7]
- Introduced a novel ***dynamic food chain model*** for three species and analyzed its long-term behavior. [8]
- Predicted ***chemical compound toxicity*** using in-vitro computational methods and feature engineering.

Technical Skills

Programming: Python, R, C++, Bash, MATLAB, SQL
ML Packages: PyTorch, Tensorflow, wandb, HuggingFace
Tools: Jupyter, LaTeX, FIJI, Microsoft 365, Adobe Illustrator, Docker
Concepts: Machine Learning, Computer Vision, Foundation Models, Self-supervised Learning, Multimodal Learning, Temporal Modeling for Video, Video Understanding, 3D Segmentation, Low-latency Inference, Diffusion, Latent Flow Matching, Distributed Training, Model Quantization, Domain Adaptation, Computational Pathology, Medical Image Analysis, Time Series Forecasting, Mathematical Modeling
Math Concepts: Data Analysis, Numerical Methods, Real Analysis, Modern Algebra, Number Theory, Statistics, Probability

Publications

First Author

[1] BM1B: Data-Efficient Scaling of Bone Marrow Foundation Models through Domain-Specific Self-Supervised Learning and Dense Morphology-Preserving Representation Learning — *MedFMB @ ECCV, 2026*
[2] Sparsely Supervised Surgical Video Segmentation with Reliable Asymmetric Dual Memory – MICCAI, 2026
[3] When Swin Transformer Meets KANs: An Improved Transformer Architecture for Medical Image Segmentation — *IEEE ISBI, 2026*
[4] Universal Conditional Networks for Cartilage Segmentation with Sparsely Annotated Data — *Nature Scientific Reports, 2025*
[5] ConUNETR: A Conditional Transformer Network for 3D Micro-CT Embryonic Cartilage Segmentation — *IEEE ISBI, 2024*
[6] SHMC-Net: A Mask-guided Feature Fusion Network for Sperm Head Morphology Classification — *IEEE ISBI, 2024*
[7] Probabilistic Analysis of Revenues in Online Games — *University of Southern Mississippi, 2020*
[8] Hunting Co-operation in the Middle Predator in Three Species Food Chain Model — *MAA Proceedings, 2020*

Co-authored

[9] Lymphovision: A lymphoma-specialized foundation model for histology-based lymphoma classification and subtyping — *Blood, 2025*
[10] A Point in the Right Direction: Vector Prediction for Spatially-aware Self-supervised Volumetric Rep. Learning — *IEEE ISBI, 2023*
[11] Keep Your Friends Close & Enemies Farther: Debiasing Contrastive Learning with Spatial Priors in 3D Images — *IEEE BIBM, 2022*
[12] Unsupervised Feature Clustering Improves Contrastive Representation Learning for Medical Image Segmentation — *IEEE BIBM, 2022*
[13] SwIPE: Efficient and Robust Medical Image Segmentation with Implicit Patch Embeddings — *MICCAI, 2023*
[14] H-CNN-ViT: A Hierarchical Gated Attention Multi-Branch Model for Bladder Cancer Recurrence Prediction — *IEEE BIBM, 2025*
[15] IHCSurv: Effective Immunohistochemistry Priors for Cancer Survival Analysis in Gigapixel Multi-stain WSIs — *MICCAI, 2024*
[16] Path-GPTOmic: A Balanced Multi-modal Learning Framework for Survival Outcome Prediction — *IEEE ISBI, 2024*
[17] Boosting Medical Image Classification with Segmentation Foundation Model — *IEEE ISBI, 2024*
[18] Embryonic Cranial Cartilage Defects in the Fgfr$^{3Y367C/+}$ Mouse Model of Achondroplasia — *Anatomical Record, 2023*

Scholarships, Grants, Honors, and Achievements

2024 IEEE International Symposium on Biomedical Imaging (ISBI2024) **Travel grant** ($800)	ISBI 2024
Graduate School **Professional Development Fund** ($1,250) and **Conference Presentation Grant** ($450)	UND 2024
CSE Select **Fellowship** Award (*1/40 incoming Ph.D students; yearly stipend worth* $40,000)	UND 2020-2025
Wright W. and Annie R. Cross **Endowment** ($10,500) and Danny R. Carter Endowed **Scholarship** ($4,000)	USM 2017-2020
Eagle SPUR **grant** for Undergraduate Research ($2,000) and Honors Keystone **Scholarship** ($2,000)	USM 2019
First Place, Mathematics Comprehensive Exam (MFT)	USM 2019
Second Runner Up: Best Undergraduate Paper	MAA Meeting 2019
Finalist, Integration Bee	MAA Meeting 2018
Nominated for College of Science and Technology's **Outstanding Sophomore Award**	USM 2017
Burner Science & Tech. **Scholarship** ($800), Wallace C. & Lynn L. Pye Endowed **Scholarship** ($800)	USM 2017

Teaching Experiences

The University of Notre Dame — Notre Dame, IN
Graduate Teaching Assistant — *08/2020 – 12/2025*

- **Design & Analysis of Algorithms**; **Complexity & Algorithms**; **Mobile App. Design**; **Discrete Mathematics**
- Prepared lecture slides, graded submissions, created answer keys, and held office hours.

STEM Project Leader | Warrior-Scholar Project — *06/2023, 06/2024*

- **Medical Image Analysis**: Designed and conducted a Bootcamp to prepare veterans for undergraduate research.
- **Introduction to Data Science**: Conducted a Bootcamp to prepare veterans for undergraduate coding classes.

Students Mentored

- **Haoyan Shi** (Undergraduate Intern, 2025) – *Published author, Currently a PhD student at University of Notre Dame*
- **Zihao Zhao** (Undergraduate Intern, 2024) – *Currently a PhD student at Rutgers*
- **Maria Jose Gomez** (High School Intern, 2024) – *Published author, applying to undergraduate programs*
- **Sirui Li** (Undergraduate Intern, 2023) – *Currently a PhD student at UCLA*
- **Santiago Rodriguez** (Undergraduate Intern, 2023) – *Now a Software Engineer at Apple*